

# H D M - 4

HIGHWAY DEVELOPMENT & MANAGEMENT

## volume **two** Applications Guide

Henry G. R. Kerali

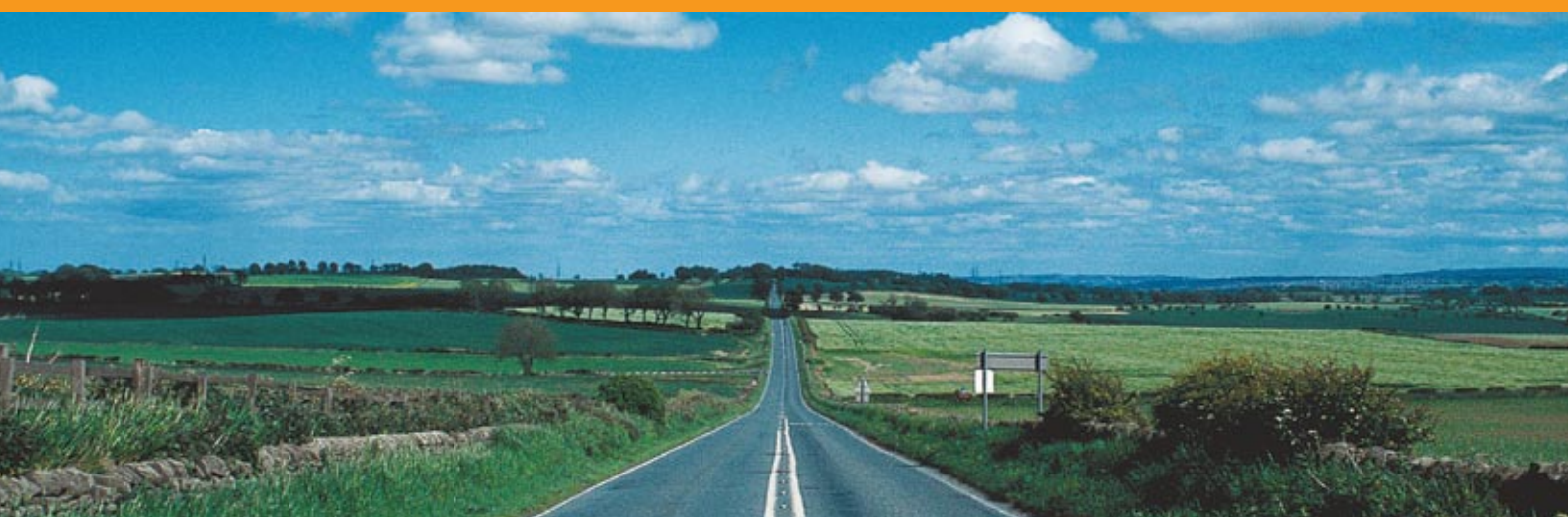
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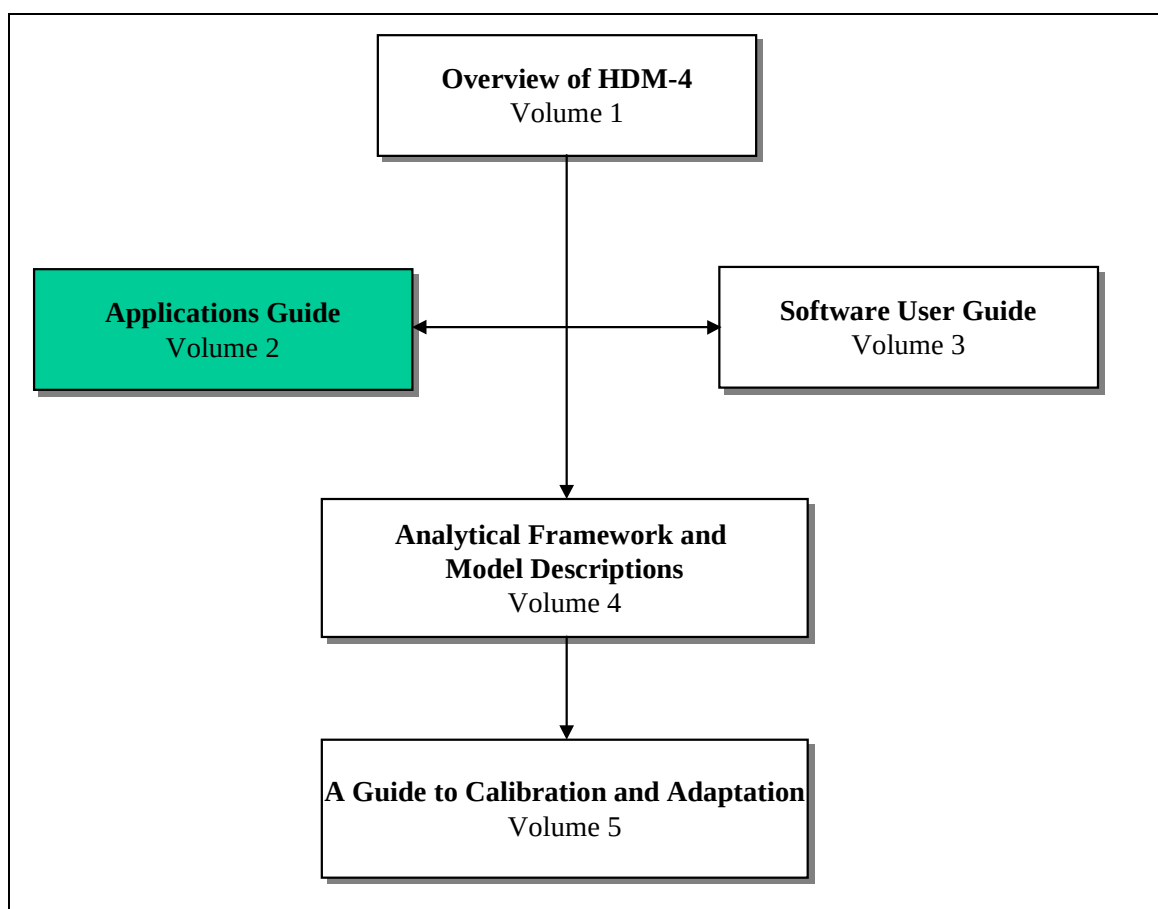
THE HIGHWAY DEVELOPMENT AND MANAGEMENT SERIES

# About This Manual

This Version 1.0 edition of the [Applications Guide](#) is a task-oriented guide describing typical examples of different types of analyses.

It is to be used by the frequent user who wishes to know how to perform a task or create a study.

The [Applications Guide](#) is one of five manuals comprising the suite of HDM-4 documentation (see Figure 1).



**Figure 1 HDM-4 documentation suite**

The suite of documents comprise:

- **Overview of HDM-4 (Volume 1)**

A short executive summary describing the HDM-4 system. It is intended to be used by all readers new to HDM-4, particularly high level management within a road organisation.

- **Applications Guide (Volume 2)**

A task oriented guide describing typical examples of different types of analyses. It is to be used by the frequent user who wishes to know how to perform a task or create a study.

- **Software User Guide (Volume 3)**

Describes the HDM-4 software. It is a general purpose document which provides an understanding of the software user interface.

- **Analytical Framework and Model Descriptions (Volume 4)**

Describes the analytical framework and the technical relationships used within the HDM-4 model. It contains very comprehensive reference material describing, in detail, the characteristics of the modelling and strategy incorporated in HDM-4. It is to be used by specialists or experts whose task is to carry out a detailed study for a road management organisation.

- **A Guide to Calibration and Adaptation (Volume 5)**

Suggests methods for calibrating and adapting HDM models (as used in HDM-III and HDM-4), to allow for local conditions existing in different countries. It discusses how to calibrate HDM-4 through its various calibration factors. It is intended to be used by experienced practitioners who wish to understand the detailed framework and models built into the HDM-4 system.

**Notes:**

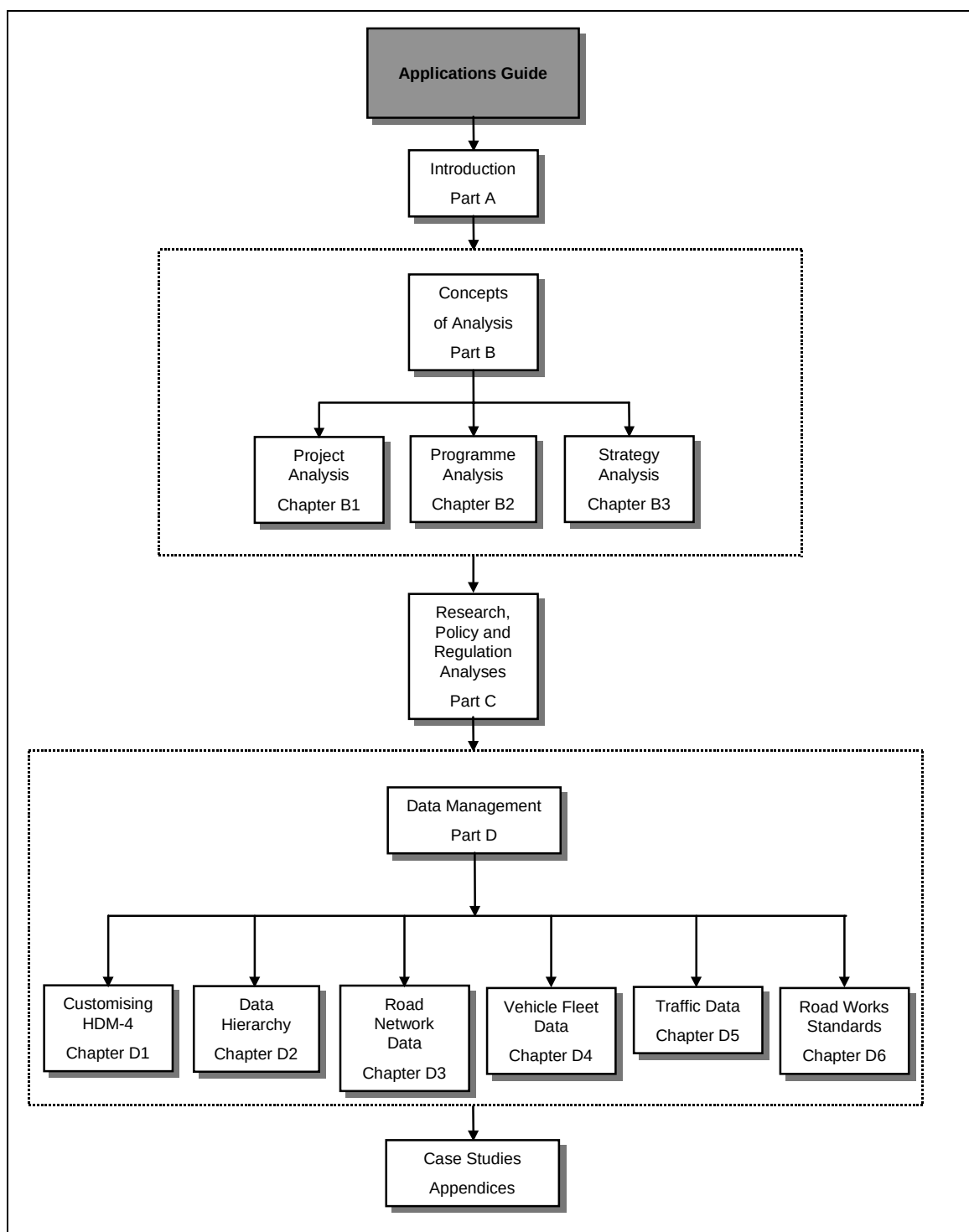
- 1 Volumes 1, 2 and 3 are designed for the general user.
- 2 Volumes 4 and 5 will be of greatest relevance to experts who wish to obtain low level technical detail. However, Volume 5, in particular, presents very important concepts, which will be of interest to all users.

## Structure of the ‘Applications Guide’

The aim of this [Applications Guide](#) is to demonstrate the different applications of HDM-4 through selected case studies. The case studies are described in detail and may be reviewed using the data files included on the CD-ROM.

The information in the [Applications Guide](#) is structured in four parts (see Figure 2).

Part A contains an overview of HDM-4, with a short description of the economic analysis framework used, and the HDM-4 Technical Models are briefly described. Part B describes the concepts of analysis for projects, programmes and strategies. Part C describes typical methods of using HDM-4 for Research, Policy and Regulation analyses. Details of the data requirements for HDM-4 are given Part D. Examples of case studies are given in the Appendices.



**Figure 2 Schematic layout of the Applications Guide documentation**

A brief description of the four parts follows:

■ **Part A - Introduction**

Provides an overview of HDM-4, with a short description of the economic analysis framework used.

■ **Part B - Concepts of analysis**

Describes the concepts used for the following analyses:

□ **Project Analysis**

Describes the overall procedure for project analysis in HDM-4. This allows users to assess the physical, functional and economic feasibility of specified project alternatives by comparison against a base case (**without-project**) alternative.

□ **Programme Analysis**

Describes the concept of Programme Analysis. This deals with the preparation of a multi-year rolling programme for a road network in which candidate investment options are identified and selected, subject to resource constraints. Road networks are analysed section by section and estimates are produced of road works and expenditure requirements for each road section for each year of the funding period.

□ **Strategy Analysis**

Describes the procedure for Strategy Analysis. This is concerned with the analysis of a chosen network as a whole, for preparing long range planning estimates of expenditure needs for road development and conservation under different budget scenarios. The road network is characterised by lengths of road in different categories defined by parameters such as road class, surface type, and pavement condition or traffic flow. Estimates are produced of expenditure requirements for medium to long term periods of 5-40 years.

■ **Part C - Research, Policy and Regulation Analyses**

Describes typical applications of HDM-4 for Research, Policy and Regulation analyses.

■ **Part D - Data Management**

Data Management comprises the following chapters:

Chapter D1 - Customising HDM-4

Chapter D2 - Data Hierarchy

Chapter D3 - Road Transport

Chapter D4 - Vehicle Fleet Data

Chapter D5 - Traffic Data

Chapter D6 - Road Works Standards

■ **Appendices - Case studies**

Case studies for Project, Programme and Strategy analyses are supplied as appendices. The [Applications Guide](#) will be updated with new case studies as future versions of HDM-4 are released.

## ISOHDM Products

The products of the International Study of Highway Development and Management Tools (ISOHDM) consist of the HDM-4 suite of software, associated example case study databases, and the Highway Development and Management Series collection of guides and reference manuals. This Volume is a member of that document collection.

## Customer contact

Should you have any difficulties with the information provided in this suite of documentation please do not hesitate to report details of the problem you are experiencing. You may send an E-mail or an annotated copy of the manual page by fax to the number provided below.

The ISOHDM Technical Secretariat welcomes any comments or suggestions from users of HDM-4. Comments on the [Applications Guide](#) should be sent to the following address:

E-mail: [isohdm@bham.ac.uk](mailto:isohdm@bham.ac.uk)  
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## Change details

This is the first formal edition (Version 1.0) of the HDM-4 documentation.

## Related documentation

### HDM-4 documents

The Highway Development and Management Series Collection is ISBN: 2-84060-058-7, and comprises:

Volume 1 - [Overview of HDM-4](#), ISBN: 2-84060-059-5

Volume 2 - [Applications Guide](#), ISBN: 2-84060-060-9

Volume 3 - [Software User Guide](#), ISBN: 2-84060-061-7

Volume 4 - [Analytical Framework and Model Descriptions](#), ISBN: 2-84060-062-5

Volume 5 - [A Guide to Calibration and Adaptation](#), ISBN: 2-84060-063-3

## Future documentation

The following documents will be issued at a later release:

Volume 6 - Modelling Road Deterioration and Works Effects, ISBN: 2-84060-102-8

Volume 7 - Modelling Road User and Environmental Effects, ISBN: 2-84060-103-6

## Terminology handbooks

*PIARC Lexicon of Road and Traffic Engineering* - First edition. Permanent International Association of Road Congresses (PIARC), Paris 1991. ISBN: 2-84060-000-5

*Technical Dictionary of Road Terms* - Seventh edition, English - French. PIARC Commission on Terminology, Paris 1997. ISBN: 2-84060-053-6

## General reference information

Further details on HDM-4 may be obtained from the following:

- ISOHDM Technical Secretariat  
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- ISOHDM Project Co-ordinator  
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- **Asian Development Bank (ADB)**
- **Department for International Development (DFID) in the United Kingdom**
- **Swedish National Road Administration (SNRA)**
- **The World Bank**

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- **Finnish National Road Administration (Finnra)**
- **Inter-American Federation of Cement Producers (FICEM)**

Many other organisations and individuals in a number of countries have also contributed in terms of providing information, or undertaking technical review of products being produced.

The study has been co-ordinated by the ISOHDM Technical Secretariat at the University of Birmingham in the United Kingdom. A number of organisations participated in the research including:

- **Finnra**  
Specification of the strategic and programme analysis applications.
- **FICEM**  
Development of deterioration and maintenance relationships for Portland cement concrete roads.
- **The Highway Research Group (HRG)**, School of Civil Engineering, The University of Birmingham  
Responsible for system design and software development.
- **Road Research Institute (IKRAM)** in Malaysia supported by N.D. Lea International (NDLI)  
Responsible for providing updated relationships for road deterioration and road user costs.
- **Transport Research Laboratory (TRL)** in the United Kingdom  
Responsible for review and update of flexible pavement deterioration relationships.
- **SNRA**  
Responsible for developing deterioration relationships for cold climates, road safety, environmental effects, and supporting HRG with system design.

All research organisations received support from local and regional staff, visiting experts and external advisers, to ensure that a high standard of quality and international consensus was achieved. A number of other countries and individuals have supported this work through supplying expert advice and reviewing the products.

## Copyright statement

These HDM-4 products have been produced by the International Study of Highway Development and Management Tools (ISOHDM), sponsored by The World Bank, the Asian Development Bank, the Department for International Development (UK), the Swedish National Road Administration, and other sponsors. The HDM-4 products are jointly published by The World Road Association (PIARC), Paris and The World Bank, Washington, DC.

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This copyright covers all documents and document components, computer software and data delivered as components of the HDM-4 product, in any physical or electronic forms.



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### Volume **one**: Overview of HDM-4

A short executive summary describing the HDM-4 system. It is intended to be used by all readers new to HDM-4, particularly high-level management within a road organisation.  
ISBN: 2-84060-059-5

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ISBN: 2-84060-063-3

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This volume describes the technical background of theory, observation, knowledge and international practice in which the HDM-4 models for road deterioration and works effects have been developed.  
ISBN-2-84060-102-8

### Volume **seven**: Modelling Road User and Environmental Effects

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ISBN-2-84060-103-6

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## Highway Development and Management Series

The Highway Development and Management system (HDM-4) provides a harmonised systems approach to road management, with adaptable and user-friendly software tools. It is a powerful tool for conducting project appraisals and analyses of road management and investment alternatives.

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